



NARAYANA
IIT-JEE ACADEMY-INDIA

JR STAR CO SC(MODEL-A&B)

Duration: 1 Hr

Date: 18.07.2024

Max. Marks: 200

CHEMISTRY NCERT TEST – 3

General Instructions:

- Attempt all the questions, each correct answer carries +4 and wrong answer -1.

SYLLABUS: THERMODYNAMICS

- Thermodynamics is not concerned about _____.
 - energy changes involved in a chemical reaction.
 - the extent to which a chemical reaction proceeds.
 - the rate at which a reaction proceeds.
 - the feasibility of a chemical reaction.
- Which of the following statements is correct?
 - The presence of reacting species in a covered beaker is an example of open system.
 - There is an exchange of energy as well as matter between the system and the surroundings in a closed system.
 - The presence of reactants in a closed vessel made up of copper is an example of a closed system.
 - The presence of reactants in a thermos flask or any other closed insulated vessel is an example of a closed system.
- The volume of gas is reduced to half from its original volume. The specific heat will be _____.
 - reduce to half
 - be doubled
 - remain constant
 - increase four times
- $\Delta_f U^\circ$ of formation of $CH_4(g)$ at certain temperature is $-393 kJ mol^{-1}$. The value of $\Delta_f H^\circ$ is
 - zero
 - $< \Delta_f U^\circ$
 - $> \Delta_f U^\circ$
 - equal to $\Delta_f U^\circ$
- In an adiabatic process, no transfer of heat takes place between system and surroundings. Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following
 - $q = 0, \Delta T \neq 0, w = 0$
 - $q \neq 0, \Delta T = 0, w = 0$
 - $q = 0, \Delta T = 0, w = 0$
 - $q = 0, \Delta T < 0, w \neq 0$

- The pressure-volume work for an ideal gas can be calculated by using the expression $w = - \int_{V_i}^{V_f} p_{ex} dV$.

The work can also be calculated from the pV– plot by using the area under the curve within the specified limits. When an ideal gas is compressed (a) reversibly or (b) irreversibly from volume V_i to V_f . choose the correct option.

- w (reversible) = w (irreversible)
- w (reversible) < w (irreversible)
- w (reversible) > w (irreversible)
- w (reversible) = w (irreversible) + $p_{ex} \Delta V$

7. The entropy change can be calculated by using the expression $\Delta S = \frac{q_{rev}}{T}$. When water freezes in a glass beaker, choose the correct statement amongst the following :
- 1) ΔS (system) decreases but ΔS (surroundings) remains the same
 - 2) ΔS (system) increases but ΔS (surroundings) decreases.
 - 3) ΔS (system) decreases but ΔS (surroundings) increases.
 - 4) ΔS (system) decreases and ΔS (surroundings) also decreases.
8. On the basis of thermochemical equations (a), (b) and (c), find out which of the algebraic relationships given in options (i) to (iv) is correct.
- (a) $C(\text{graphite}) + O_2(g) \rightarrow CO_2(g); \Delta_r H = x \text{ kJ mol}^{-1}$
- (b) $C(\text{graphite}) + \frac{1}{2} O_2(g) \rightarrow CO(g); \Delta_r H = y \text{ kJ mol}^{-1}$
- (c) $CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g); \Delta_r H = z \text{ kJ mol}^{-1}$
- 1) $z = x + y$
 - 2) $x = y - z$
 - 3) $x = y + z$
 - 4) $y = 2z - x$
9. Enthalpy of sublimation of a substance is equal to
- 1) enthalpy of fusion + enthalpy of vapourisation
 - 2) enthalpy of fusion
 - 3) enthalpy of vapourisation
 - 4) twice the enthalpy of vapourisation
10. The spontaneity means, having the potential to proceed without the assistance of external agency. The processes which occur spontaneously are
- (a) flow of heat from colder to warmer body.
 - (b) gas in a container contracting into one corner.
 - (c) gas expanding to fill the available volume.
 - (d) burning carbon in oxygen to give carbon dioxide.
- 1) a & b only
 - 2) b & c only
 - 3) c & d only
 - 4) a, b, c & d
11. A sample containing 1.0 mol of an ideal gas is expanded isothermally and reversibly to ten times of its original volume, in two separate experiments. The expansion is carried out at 300 K and at 600 K respectively. Choose the correct option.
- (a) Work done at 600 K is 20 times the work done at 300 K.
 - (b) Work done at 300 K is twice the work done at 600 K.
 - (c) Work done at 600 K is twice the work done at 300 K.
 - (d) $\Delta U = 0$ in both cases
- 1) a & c only
 - 2) c & d only
 - 3) b & d only
 - 4) a, b, c & d

12. Match the column I with column II:

Column I

- (i) Entropy of vapourisation
- (ii) K for spontaneous process
- (iii) Crystalline solid state
- (iv) ΔU in adiabatic expansion of ideal gas

Column II

- (a) decreases
- (b) is always positive
- (c) lowest entropy
- (d) $\frac{\Delta H_{vap}}{T_b}$

1) (i)-a, (ii)-b, (iii)-d, (iv)-c

2) (i)-b,d, (ii)-b, (iii)-c, (iv)-a

3) (i)-d, (ii)-c, (iii)-b, (iv)-a

4) (i)-a, (ii)-b, (iii)-c, (iv)-d

13. **Assertion (A):** Combustion of all organic compounds is an exothermic reaction.

Reason (R) : The enthalpies of all elements in their standard state are zero.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true but R is not the correct explanation of A.
- 3) A is true but R is false.
- 4) A is false but R is true.

14. **Assertion (A) :** Spontaneous process is an irreversible process and may be reversed by some external agency.

Reason (R) : Decrease in enthalpy is a contributory factor for spontaneity.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true but R is not the correct explanation of A.
- 3) A is true but R is false.
- 4) A is false but R is true.

15. For the reaction, $X_2O_{4(l)} \rightarrow 2XO_{2(g)}$

$\Delta H = 3.3 \text{ kcal}$, $\Delta S = 20 \text{ cal K}^{-1}$ at 300 K. Hence: ΔG is

- 1) 2.7 kcal
- 2) -2.7 kcal
- 3) 9.3 kcal
- 4) -9.3 kcal

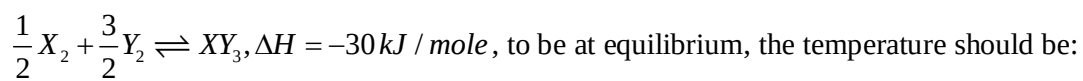
16. If the enthalpy change for the transition of liquid to vapour is 300 kJ mol^{-1} at 27°C , the entropy change for the process would be

- 1) $1000 \text{ JK}^{-1} \text{ mole}^{-1}$
- 2) $10 \text{ JK}^{-1} \text{ mole}^{-1}$
- 3) $100 \text{ JK}^{-1} \text{ mole}^{-1}$
- 4) $0.1 \text{ JK}^{-1} \text{ mole}^{-1}$

17. In which of the following conditions, the reaction is spontaneous at all temperatures.?

	ΔH	ΔS	ΔG
1)	-	-	-
2)	+	-	+
3)	+	+	+
4)	-	+	-

18. Standard entropies of X_2 , Y_2 and XY_3 are 60, 40 and 50 $JK^{-1}mol^{-1}$ respectively. For the reaction



- 1) 500 K 2) 750 K 3) 1000 K 4) 1250 K
19. Which of the following motions of molecule become more vigorous with increase in temperature due to which an increase in entropy is observed?

- 1) Translational motion 2) Rotational motion
3) Vibrational motion 4) All of these

20. Statement-I: A spontaneous process involves decrease of free energy.

Statement-II: Spontaneous reactions are always exothermic.

- 1) Both Statement-I and Statement-II are correct
2) Both Statement-I and Statement-II are incorrect
3) Statement-I is correct & Statement -II is Incorrect
4) Statement-I is incorrect & Statement -II is correct

21. Standard enthalpy is zero for

- 1) $Br_2(g)$ 2) $NH_3(g)$ 3) $O_2(g)$ 4) $CaCO_3(s)$

22. In which of the following process, Enthalpy change cannot be called as Enthalpy of atomization ?

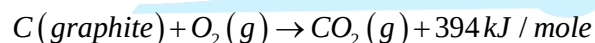
- 1) $Cu(s) \rightarrow Cu(g)$ 2) $O_2(g) \rightarrow 2O(g)$
3) $I_2(s) \rightarrow I_2(g)$ 4) $CH_4(g) \rightarrow C(g) + 4H(g)$

23. $2HCl(g) \rightarrow H_2(g) + Cl_2(g); \Delta H = +y KJ$

From the above data, calculate the heat of formation of $HCl(g)$ in kJ/mole.

- 1) $\frac{y}{2}$ 2) y 3) $-\frac{y}{2}$ 4) $-y$

24. Consider the equation,



Incorrect statement about the reaction is

- 1) It is an endothermic reaction
2) $\Delta_f^\circ H$ of carbondioxide is $-394 kJ/mol$
3) All the reactants and products are in their standard state
4) Enthalpy of combustion of graphite is $-394 KJ/mole$

25. **Statement I:** Standard enthalpy of monoclinic sulphur is zero.

Statement II: A thermo chemical equation can be balanced using fractions.

- 1) Statements I and II are correct
2) Statement I is correct and statement II is incorrect
3) Statement I is incorrect and statement II is correct

4) Statements I and II are incorrect

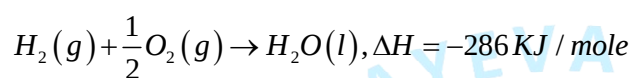
26. Enthalpy of combustion of diamond is -395 kJ/mole and Enthalpy of combustion of graphite is -393 kJ/mole . Enthalpy change when one mole of diamond converts to one mole of graphite is

1) 2 kJ 2) -2 kJ 3) 4 kJ 4) -4 kJ

27. Enthalpy of solution of $\text{MCl}_2(s)$ is -108 kJ/mole . Enthalpy of solution of $\text{MCl}_2 \cdot 6\text{H}_2\text{O}(s)$ is -81 kJ/mole . Enthalpy of hydration of $\text{MCl}_2(s)$ into its hexahydrate is

1) $+190 \text{ kJ}$ 2) $+27 \text{ kJ}$ 3) -190 kJ 4) -27 kJ

28. $\text{H}_2(g) + \frac{1}{2}\text{O}_2(g) \rightarrow \text{H}_2\text{O}(g), \Delta H = -245 \text{ kJ/mole}$



Enthalpy of condensation of water vapour

1) -41 kJ/mole 2) $+531 \text{ kJ/mole}$ 3) $+41 \text{ kJ/mole}$ 4) -531 kJ/mole

29. **Assertion (A) :** Enthalpy of neutralisation of strong acid and strong base is always a constant.

Reason (R) : The net reaction involving one gram equivalent of strong acid and one gram equivalent of strong base is always $\text{H}^+(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l)$

- 1) Both(A) and (R) are true and (R) is the correct explanation of (A)
 2) Both(A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false
 4) Both(A) and (R) are false

30. What happens when methane undergoes combustion in system A and B respectively?

Adiabatic
system
System A

Diathermic
container
System B

- 1) System A – Temperature rises, System B – Temperature remains same
 2) System A – Temperature falls, System B – Temperature rises
 3) System A – Temperature falls, System B – Temperature remains same
 4) System A – Temperature remains same, System B – Temperature rises

31. Among the following, the set of parameters that represents path functions, is

(A) $q + w$ (B) q (C) w (D) $H - TS$

- 1) (B) and (C) only 2) (B), (C) and (D) only
 3) (A) and (D) only 4) (A), (B) and (C) only

32. Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas? (Assume not expansion work is zero)

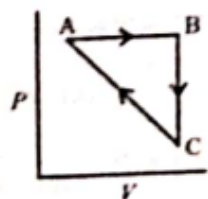
1) Cyclic process: $q = -w$

2) Adiabatic process: $\Delta U = -w$

3) Isochoric process: $\Delta U = q$

4) Isothermal process: $q = -w$

33. An ideal gas undergoes a cyclic process as shown in figure.



$$\Delta U_{BC} = -5 \text{ kJ mol}^{-1}, q_{AB} = 2 \text{ kJ mol}^{-1}$$

$$W_{AB} = -5 \text{ kJ mol}^{-1}, W_{CA} = 3 \text{ kJ mol}^{-1}$$

Heat absorbed by the system during process CA is

1) -5 kJ mol^{-1}

2) $+5 \text{ kJ mol}^{-1}$

3) 18 kJ mol^{-1}

4) -18 kJ mol^{-1}

34. An ideal gas is allowed to expand both reversibly and irreversibly in an isolated system. If T_i is the initial temperature and T_f is the final temperature, which of the following statements is correct?

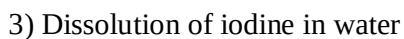
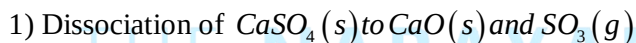
1) $(T_f)_{\text{rev}} = (T_f)_{\text{irrev}}$

2) $T_f = T_i$ for both reversible and irreversible processes

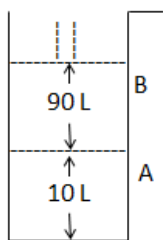
3) $(T_f)_{\text{irrev}} > (T_f)_{\text{rev}}$

4) $T_f > T_i$ for reversible process but $T_f = T_i$ for irreversible process

35. The process with negative entropy change is:



36.



Consider the figure provided:

1 mol of an ideal gas is kept in a cylinder, fitted with a piston, at the position A, at 18°C . If the piston is moved to position B, keeping the temperature unchanged, then 'x' L atm work is done in this reversible process.

$x = \underline{\hspace{1cm}}$ L atm (nearest integer)

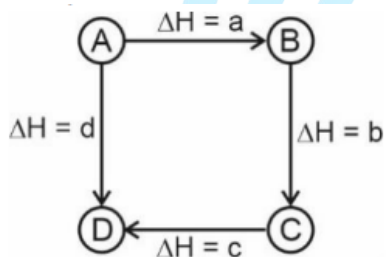
[Given: Absolute temperature = $^{\circ}\text{C} + 273.15$, $R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$]

- 1) 34 2) 55 3) 20 4) 110

37. The amount of heat required to raise the temperature of 21 g of aluminium from 250 K to 350 K is (Specific heat of aluminium is $0.4 \text{ J } (^{\circ}\text{C})^{-1} \text{ g}^{-1}$)

- 1) 420 J 2) 1100 J 3) 840 J 4) 2400 J

38. Which among the following is a correct relation for the given process?



- 1) $a - b = d - c$ 2) $a + c = d - b$ 3) $b + c = a + d$ 4) $a - d = b + c$

39. **Assertion (A)** : ΔH is negative for irreversible free expansion of an ideal gas under isothermal condition.

Reason (R) : Irreversible free expansion of an ideal gas is spontaneous process.

In the light of the above statements, choose the correct answer from the options given below:

- 1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 2) (A) is true but (R) is false
 3) (A) is false but (R) is true
 4) Both (A) and (R) are true and (R) is the correct explanation (A)

40. ΔH_f for $\text{C}_2\text{H}_4 = 12.5 \text{ kcal mol}^{-1}$

Heat of atomisation of C = $171 \text{ kcal mol}^{-1}$

Bond energy of $\text{H}_2 = 104.3 \text{ kcal mol}^{-1}$

Bond energy C – H = $99.3 \text{ kcal mol}^{-1}$

Bond energy of C = C will be

- 1) $140.9 \text{ kcal mol}^{-1}$ 2) 76 kcal mol^{-1} 3) 86 kcal mol^{-1} 4) 41 kcal mol^{-1}

41. The value of ΔH and ΔS for the reaction, $C(\text{graphite}) + CO_2(g) \rightarrow 2CO(g)$ are 170 kJ mol^{-1} and $170 \text{ J K}^{-1} \text{ mol}^{-1}$ respectively. This reaction will be in equilibrium at

- 1) 950 K 2) 900 K 3) 1100 K 4) 1000 K

42. For the following processes

$\Delta H \text{ (kJ/mol)}$



For, $2A + 3B \rightarrow 3D + E$, ΔH in kJ mol^{-1} will be

- 1) $2z + x - 2y$ 2) $z - 2x + 3y$ 3) $-z + 2x - 2y$ 4) $z - 2x - 3y$

43. If bond energy of 'P – P', 'H – H' and P – H bonds are $a \text{ kJ mol}^{-1}$, $b \text{ kJ mol}^{-1}$ and $c \text{ kJ mol}^{-1}$ respectively the heat of following reaction (in kJ mol^{-1}) will be,

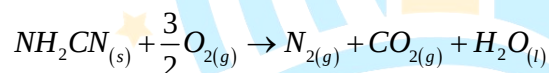


- 1) $4a + 6b - 12c$ 2) $12c + 4a - 6b$ 3) $12c + 6a - 6b$ 4) $6a + 6b - 12c$

44. The enthalpy of combustion of methane, graphite and dihydrogen at 298 K are, $-890.3 \text{ kJ mol}^{-1}$, $-393.5 \text{ kJ mol}^{-1}$, and $-285.8 \text{ kJ mol}^{-1}$ respectively. Enthalpy of formation of $CH_4(g)$ will be

- 1) $-74.8 \text{ kJ mol}^{-1}$ 2) $-52.27 \text{ kJ mol}^{-1}$ 3) $+74.8 \text{ kJ mol}^{-1}$ 4) $+52.26 \text{ kJ mol}^{-1}$

45. The reaction of cyanamide, $NH_2CN(s)$, with dioxygen was carried out in a bomb calorimeter, and ΔU was found to be $-742.7 \text{ kJ mol}^{-1}$ at 298 K. Calculate enthalpy change for the reaction at 298K.



- 1) $-742.7 \text{ K.J mol}^{-1}$ 2) $-741.5 \text{ K.J mol}^{-1}$ 3) $-845 \text{ K.J mol}^{-1}$ 4) $-680 \text{ K.J mol}^{-1}$

46. Calculate the enthalpy change on freezing of 1.0 mol of water at 10.0°C to ice at -10.0°C .

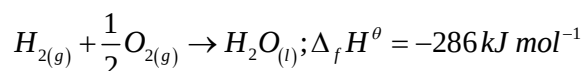
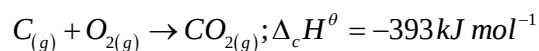
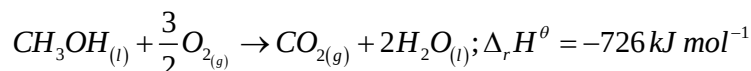
$$\Delta_{\text{fus}}H = 6.03 \text{ kJ mol}^{-1} \text{ at } 0^\circ\text{C.}$$

$$C_p[H_2O(l)] = 75.3 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$C_p[H_2O(s)] = 36.8 \text{ J mol}^{-1} \text{ K}^{-1}$$

- 1) $-7.151 \text{ KJ mol}^{-1}$ 2) -753 KJ mol^{-1} 3) -368 KJ mol^{-1} 4) $-7.53 \text{ KJ mol}^{-1}$

47. Calculate the standard enthalpy of formation of $CH_3OH(l)$ from the following data:



1) -239 KJ mol^{-1}

2) $+239 \text{ KJ mol}^{-1}$

3) -478 KJ mol^{-1}

4) $+478 \text{ KJ mol}^{-1}$

48. An ideal gas, $\bar{C}_v = \frac{5}{2}R$, is expanded adiabatically against a constant pressure of 1 atm until it doubles in volume. If the initial temperature and pressure is 298 K and 5 atm, respectively then the final temperature is _____ K (nearest interger).

[$\bar{C}_v$ is the molar heat capacity at constant volume]

1) 274

2) 300

3) 50

4) 146

49. When a 60 W electric heater is immersed in a gas for 100s in a constant volume container with adiabatic walls, the temperature of the gas rises by 5°C . The heat capacity of the given gas is _____ (Nearest integer)

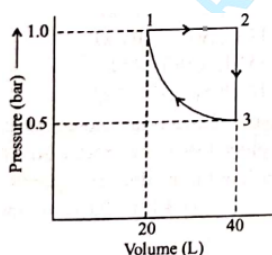
1) 900

2) 1200

3) 1500

4) 600

50. One mole of an ideal monoatomic gas is subjected to changes as shown in the graph. The magnitude of the work done (by the system or on the system) is _____ J (nearest integer)



Given: $\log 2 = 0.3, \ln 10 = 2.3$

1) 620

2) 820

3) 22.4

4) 120

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 66 YASUVA NETAAN REDDY App No. 240310228048	 69 MAJJI RISHI VARSHAN App No. 24031018100	 78 SANKAR MAHENDR KUMAR App No. 24031047378	 82 DEVANSH BATTANI App No. 240310374860 (DUP)	 84 PATHEI VISHVA PRANAV App No. 24031019438	 85 ABHIRAM CHANDRA App No. 24031006888 (DUP)	 88 P RISHAB App No. 24031069077	 90 AKSHAT CHANDRAN App No. 24031023881 (DUP)	Many More...

6
in Top 10
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7
in Top 12
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28
in Top 100
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891
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